

Corrected Mode-I Adaptive Remeshing: Localization and Fracture-Response Verification

Focused Supervisor Report – 08 October 2026

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1. Objective and decision context

The purpose of this short report is to answer one focused question: *did the corrected native Abaqus remeshing workflow localize refinement to the Mode-I crack-propagation corridor while preserving an acceptable fracture response?* The work therefore has three objectives:

1. identify why the earlier adaptive mesh refined too much of the domain;
2. demonstrate the corrected, crack-path-localized mesh; and
3. verify the corrected mesh using reaction force, initial stiffness, spatial damage, and energy evidence.

Main outcome. The earlier Step-1/pre-peak remeshing state produced broad refinement, whereas evaluation of the localized Step-2 field produced a narrow horizontal corridor. The governed ET1 corrected mesh contains 14,483 finite elements, with 64.12% of them in $|y - 0.5| \leq 0.05$ mm. It preserves the initial stiffness to -0.0261% relative to the fixed reference and predicts $F_{\max}$ within -1.8563% .

Table 1: Supervisor-facing status at the start of the 08 October meeting.

Topic	Status	Evidence-based interpretation
Earlier adaptive mesh	Superseded for localization assessment	Native remeshing worked, but the selected Step-1 field drove broad refinement.
Corrected Step-2 mesh	Verified	Refinement follows the horizontal ligament and avoids refinement on the initial slit flanks.
Mechanical response	Verified over tested cases	K_0 , $F_{\max}$, u_{peak} , and crack symmetry remain stable or acceptably close to the fixed reference.
Energetic interpretation	Qualified with limitation	Component histories are available; the exact global energy identity remains not yet closed.
Gate status	Ready for supervisor sign-off	Gate 6C, Mode-II, and Gate 7 remain on hold until a formal decision.

This report deliberately omits the detailed gate ledger, long derivations, intermediate failed packages, and full job history. Those remain in the governed project records.

2. Problem with the earlier adaptive mesh

The earlier native adaptive mesh was operational, but it was not an efficient representation of the intended fracture process zone. The 1% target-error case contained 71,320 finite elements and refined a large part of the specimen rather than concentrating resolution primarily along the future horizontal crack path. Figure 1 retains only the requested full-domain and crack-tip views of that case.

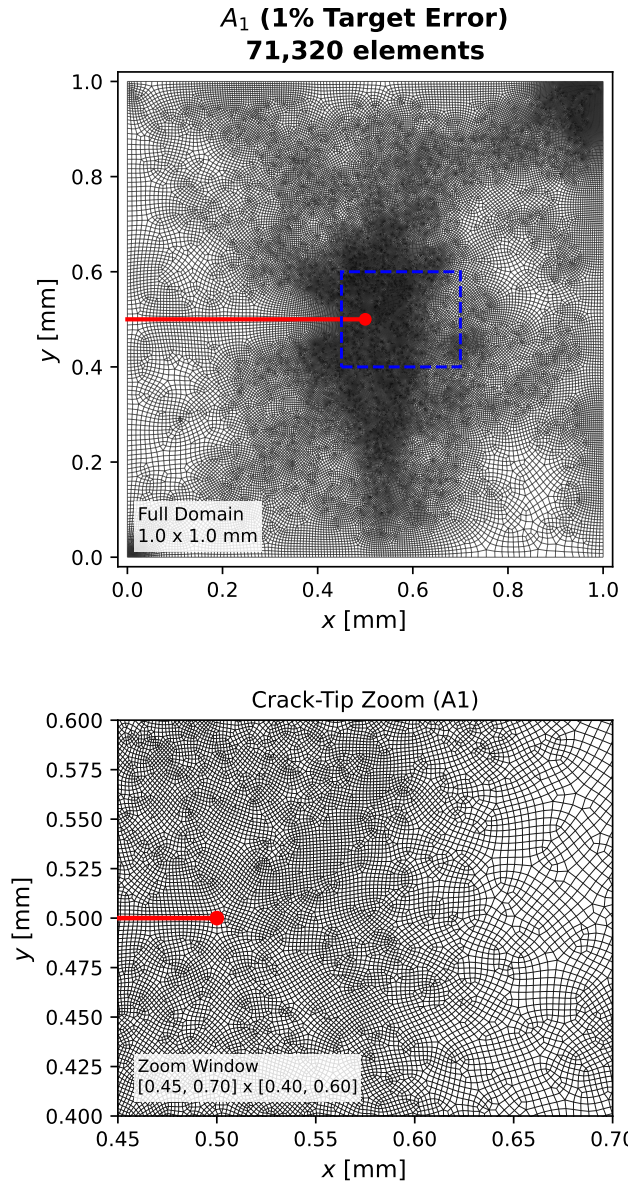


Figure 1: Earlier 71,320-element Step-1/pre-peak adaptive mesh: full-domain view (top) and crack-tip zoom (bottom). Although the crack tip is refined, fine resolution is distributed broadly through the specimen.

Scientific conclusion. Element count alone was not a sufficient quality measure. The central problem was spatial selectivity: too much resolution was placed away from the propagating ligament.

The broad pattern had two distinct issues that must not be conflated. First, an early boundary-set formatting defect omitted bottom-edge constraints and distorted stiffness; that defect was corrected and closed. Second, even with correct boundary conditions, the remeshing field had been evaluated at a Step-1/pre-peak state, before the fracture process had localized along the ligament. The second issue explains the inefficient mesh morphology addressed here.

3. Corrections introduced

The decisive change was to align the remeshing evaluation state with the physical feature that the mesh is intended to resolve. The native **RemeshingRule** now evaluates the localized Step-2 companion field instead of treating an early pre-peak error field as the final sizing basis.

Table 2: Difference table: earlier state, implemented correction, and observed effect.

Item	Earlier state / issue	Correction applied	Observed effect
Remeshing evaluation state	Step-1/pre-peak field emphasized the initial crack-tip neighborhood and broad background error.	Evaluate the localized Step-2 field after the crack-propagation pattern is established.	Refinement extends along the horizontal ligament instead of filling most of the domain.
Boundary-condition set	A long N_BOTTOM card was parsed incompletely in an early model.	Wrap node identifiers to valid Abaqus keyword lines.	All intended supports are applied; K_0 returns to the reference range.
Quality criterion	Total element count received excessive emphasis.	Judge corridor share, band width, far-field fraction, crack symmetry, and response together.	Mesh efficiency is assessed by physical localization rather than count matching alone.
Tolerance study	A single tolerance could hide morphology sensitivity.	Evaluate errorTarget $\in \{1, 2, 3, 5\}\%$ with identical bounds and region.	ET1 is target-like; ET2–ET5 progressively lose ligament resolution.
Post-processing	Results were dispersed across separate checks.	Use single-job provenance for RF- u , K_0 , energy, phase field, and crack-path metrics.	Mechanical and spatial conclusions can be traced to exact jobs without cross-copying.

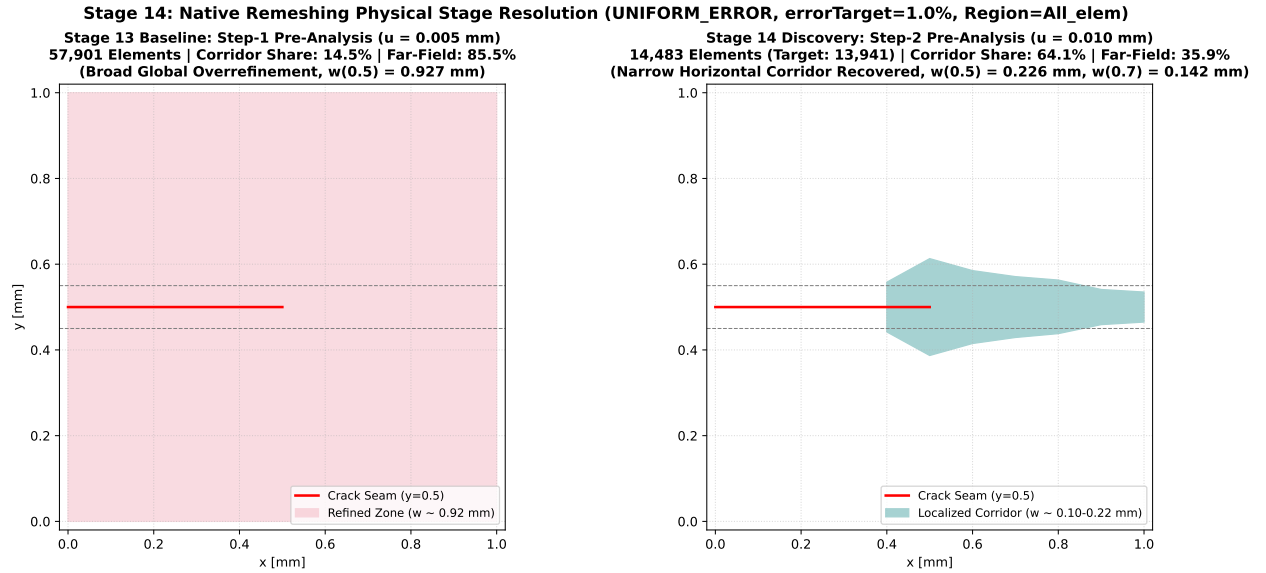


Figure 2: Isolated effect of physical-stage selection at **errorTarget=1%**. The historical Stage-13/Step-1 morphology (left) has exactly 57,901 elements (56,351 CPE4 + 1,550 CPE3; deck SHA-256 872B54A6...) and produces broad global refinement. It is distinct from the later 57,929-element spatial-fine fracture candidate. The corrected Step-2 ET1 mesh (right) has 14,483 elements, 64.1% corridor share, and 35.9% far-field share.

Scientific conclusion. The main localization improvement comes from evaluating the physically relevant Step-2 field, not from tuning the element count after the fact.

4. Corrected adaptive mesh

Figure 3 shows the governed corrected ET1 mesh at both specimen and ligament scales. It contains 14,483 finite elements (14,082 quadrilaterals and 401 triangles) and 14,456 finite-element nodes. The minimum and median area-equivalent element sizes are $0.760\ \mu\text{m}$ and $2.597\ \mu\text{m}$, respectively.

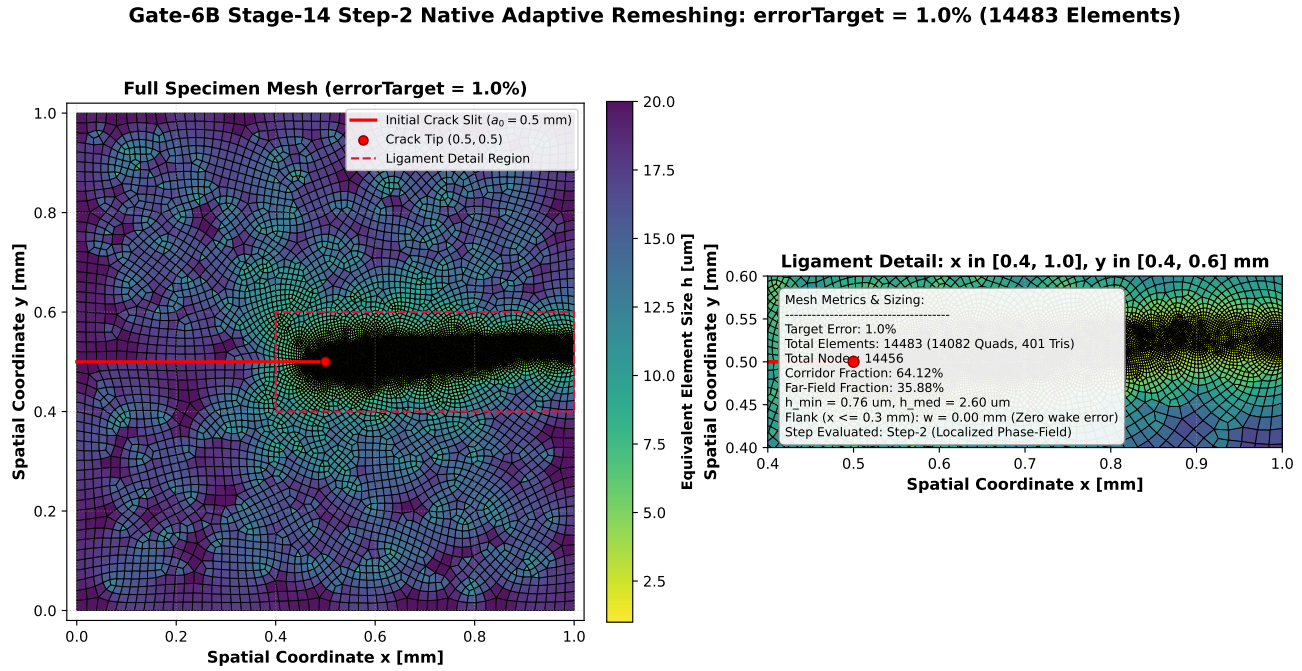


Figure 3: Corrected Step-2 ET1 adaptive mesh: full-domain view (left) and ligament zoom (right). The refined corridor begins at the initial crack tip and follows the expected horizontal propagation line. The initial slit flanks at $x \leq 0.3\ \text{mm}$ show zero artificial refinement in the governed width metric.

Scientific conclusion. The corrected mesh concentrates its finest resolution in the physically relevant crack-propagation region while retaining a comparatively coarse far field.

The morphology is qualitatively consistent with the localized adaptive pattern reported by Pandey and Kumar [2025]. This statement concerns the character of the refinement corridor, not an exact reproduction of their reported 13,941 elements. The publication does not expose enough implementation detail to establish exact count identity, and that limitation has already been accepted by the supervisor.

5. Mesh roles and tolerance sensitivity

The Step-2 error-target sweep separates localization quality from simple element reduction. All four tolerance cases use the same evaluation state, region, minimum/maximum size bounds, and coarsening policy; only the target tolerance changes.

Three mesh roles must remain separate. The historical Stage-13/Step-1 morphology has 57,901 elements and is used only to show broad, inefficient refinement. Corrected ET1 has 14,483 elements and is the preferred practical adaptive mesh. The 57,929-element spatial-fine mesh is a distinct topology and fracture solve (Job 1410504.mm~~aster~~02; deck SHA-256 537C8C66...) used to test internal adaptive spatial convergence.

Table 3: Corrected Step-2 tolerance cases and spatial-fine convergence check. Percent peak-force differences are relative to the fixed reference $F_{\max} = 0.757778$ kN.

Case	Base FE	Corridor share	K_0 [kN/mm]	$F_{\max}$ [kN]	$\Delta F_{\max}$ vs fixed	u_{peak} [mm]	Purpose
ET1 (1%)	14,483	64.12%	137.9096	0.743701	−1.8563%	0.005733	Preferred corrected adaptive mesh
ET2 (2%)	6,112	41.07%	137.9761	0.756367	−0.1862%	0.005841	Tolerance sensitivity
ET3 (3%)	5,189	34.59%	137.9775	0.759407	+0.2150%	0.005876	Tolerance sensitivity
ET5 (5%)	4,692	27.51%	138.0091	0.765400	+1.0058%	0.005926	Tolerance sensitivity
Spatial-fine	57,929	—	137.8410	0.741633	−2.1305%	0.005717	Adaptive spatial-convergence check

All corrected cases satisfy the predeclared stiffness ($|\Delta K_0| \leq 0.5\%$) and peak-force ($|\Delta F_{\max}| \leq 5\%$) acceptance ranges. The important result is that ET1 to spatial-fine refinement changes $F_{\max}$ by only 0.28%, indicating internal adaptive spatial convergence; ET1 and spatial-fine remain 1.86% and 2.13% below the independent structured fixed reference, respectively. ET2–ET5 remain useful tolerance sensitivities, but their lower counts progressively reduce process-corridor coverage.

6. Mechanical verification: reaction force and energy

Figure 4 compares exactly three scientific roles: the fixed structured reference, corrected ET1, and the 57,929-element spatial-fine adaptive candidate. The fine candidate was completed over the full $u = 0.010$ mm horizon using qualified 8-thread shared-memory execution (Job 1410504.mmmaster02).

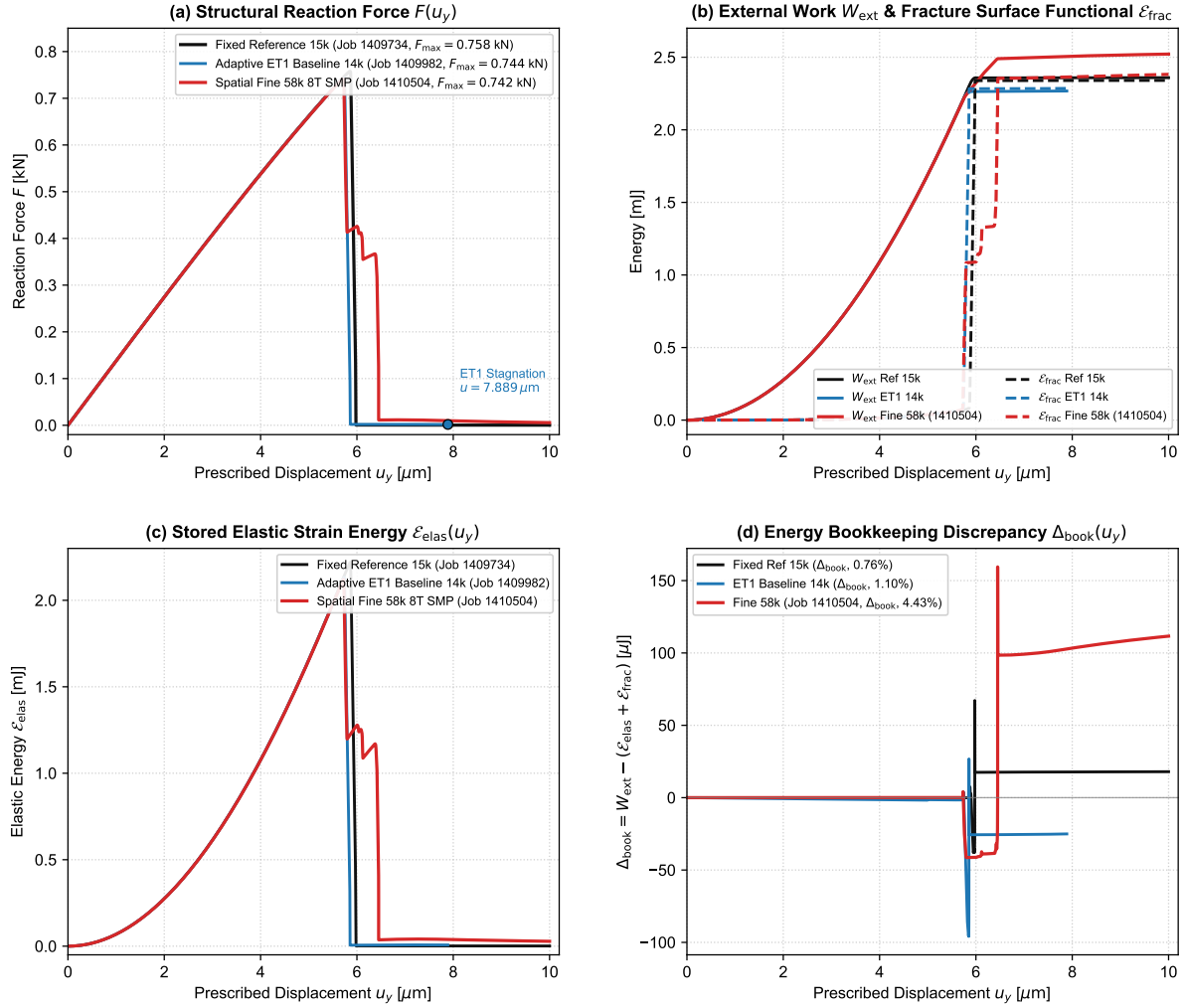


Figure 4: Three-case Gate-6B global synthesis: (a) reaction force; (b) external work and phase-field fracture-energy functional; (c) stored elastic energy; and (d) energy-bookkeeping difference. ET1 is censored at $u = 0.007889$ mm after a 98.5% load drop; no values are forward-filled.

Scientific conclusion. The corrected adaptive meshes reproduce the elastic branch and peak response. The 14.5k-to-57.9k change in $F_{\max}$ and u_{peak} is below 0.28%, establishing internal adaptive spatial convergence, while the remaining 2.13% peak offset to the structured fixed reference is unresolved.

Table 4: Compact result summary. Energies are compared only over each job's valid reached domain.

Case	Base FE	K_0 [kN/mm]	$F_{\max}$ [kN]	E_{frac} [mJ]	Energy residual $\varepsilon_{\text{book}}$ [%]	Valid domain / interpretation
Fixed reference	15,192	137.9455	0.757778	2.340220	0.7607	Full horizon, $u = 0.0100$ mm
Corrected ET1	14,483	137.9096	0.743701	2.285469	1.1048	Reached $u = 0.007889$ mm; 98.5% load drop
Spatial-fine 8T	57,929	137.8410	0.741633	2.381941	4.4263	Full horizon, $u = 0.0100$ mm

Here $\varepsilon_{\text{book}} = |W_{\text{ext}} - (E_{\text{elas}} + E_{\text{frac}})| / |W_{\text{ext}}| \times 100\%$. Thus 4.4263% is an endpoint energy-bookkeeping residual, not a fracture-energy error. The curves are physically bounded and the fine run closes the full displacement horizon, but this two-term residual is not proof of an exact joint potential for the staggered history-field algorithm. The global energy identity therefore remains formally not yet closed.

7. Initial stiffness verification (K_0)

The initial elastic branch is the cleanest test of geometry, support conditions, material constants, and reference-point coupling. The canonical ordinary-least-squares fit uses the first 400 increments up to $u \leq 0.0010$ mm.

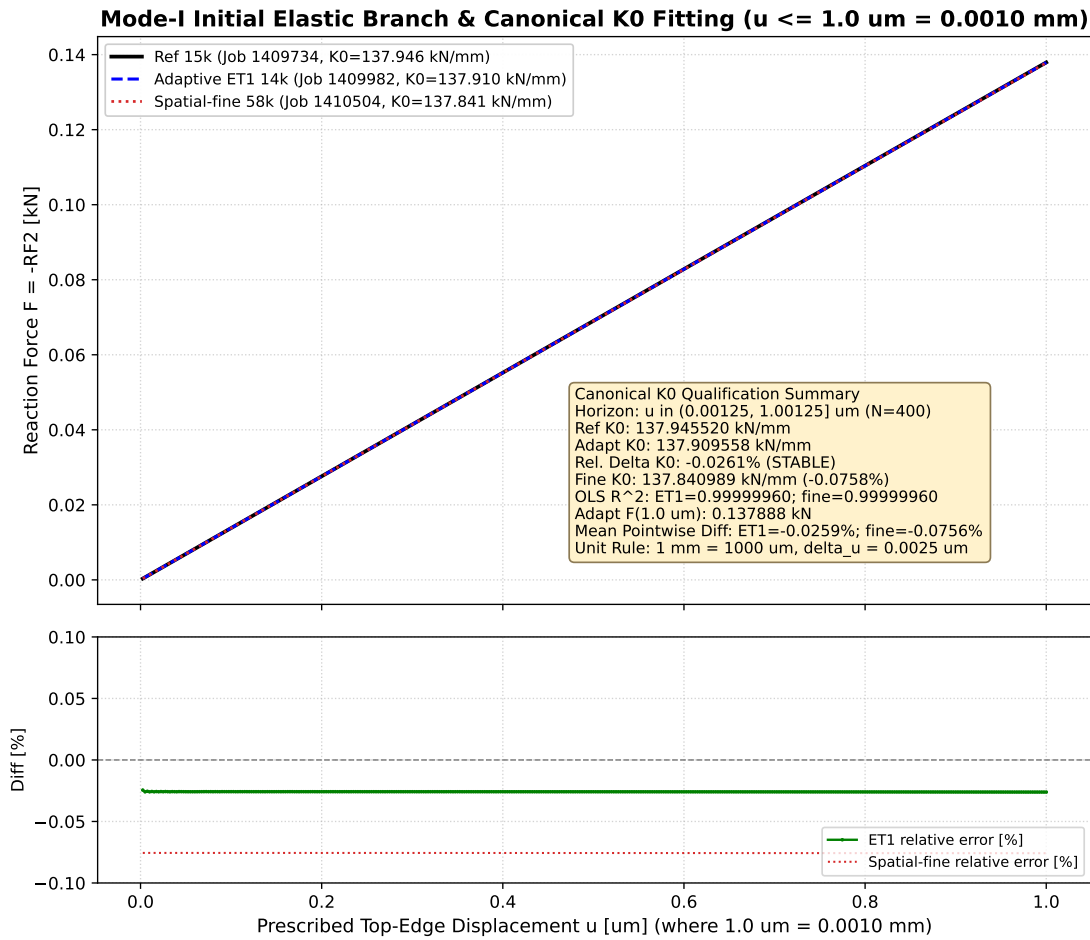


Figure 5: Canonical K_0 comparison for the fixed reference, corrected ET1, and spatial-fine adaptive mesh. The fitted stiffnesses are 137.945520, 137.909558, and 137.840989 kN/mm, respectively; both adaptive results remain within 0.08% of the reference.

Scientific conclusion. The corrected adaptive mesh preserves the initial structural response. The earlier boundary-condition defect is closed and does not explain the remaining peak-force difference.

The spatial-fine 57,929-element candidate gives $K_0 = 137.840989 \text{ kN/mm}$ (-0.0758% relative to the reference), confirming stiffness stability across a fourfold change in adaptive element count.

8. Spatial damage and crack-path verification

Global RF– u agreement is not sufficient by itself. The phase-field distribution must also remain centered on the Mode-I symmetry plane and progress along the intended ligament.

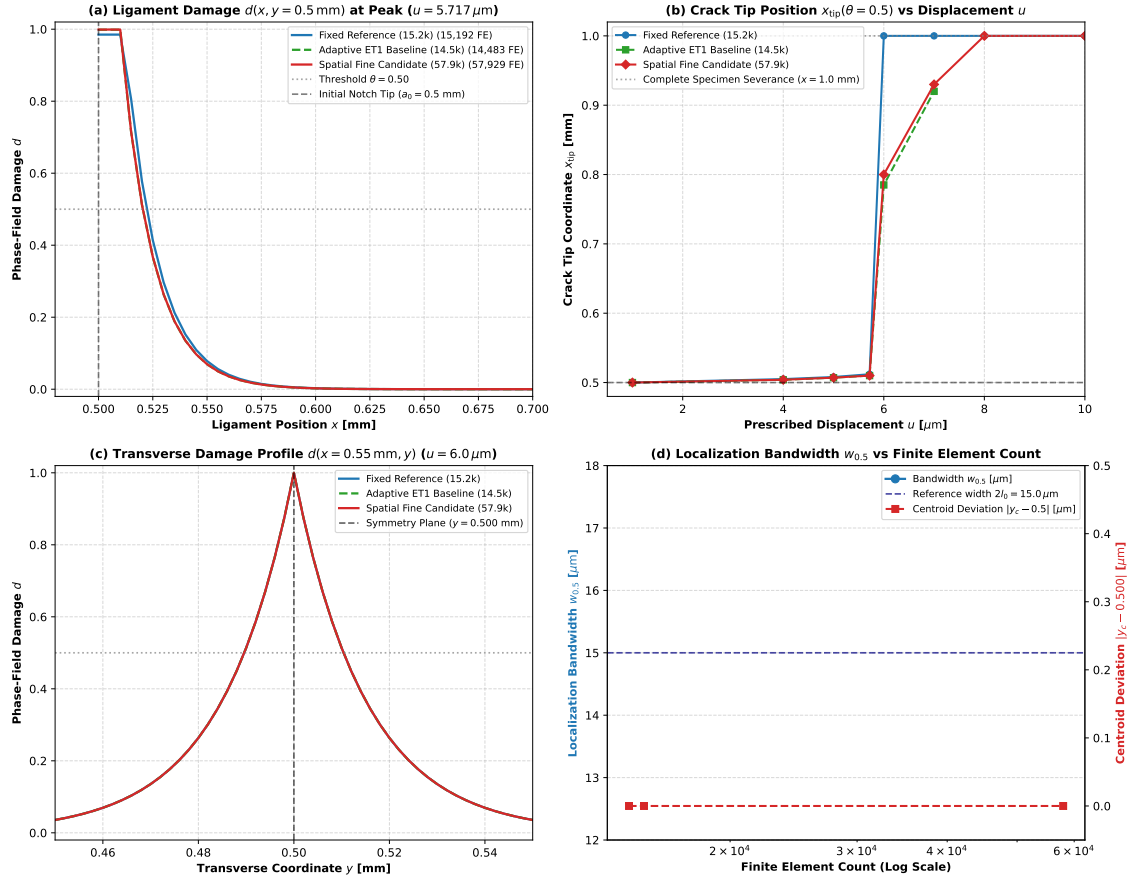


Figure 6: Three-case spatial verification for the fixed reference, corrected ET1, and spatial-fine adaptive mesh: peak ligament profiles, crack-tip motion, transverse damage profiles, localization bandwidth, and centroid deviation.

Scientific conclusion. The corrected meshes preserve a horizontal Mode-I crack path. Pre-peak ET1 and spatial-fine ligament profiles differ by at most 0.32% in the governed L_2 measure, and the transverse damage centroid remains at $y = 0.500 \text{ mm}$ within the evaluated precision.

The observed transverse localization bandwidth is approximately $w_{0.5} \approx 15 \mu\text{m}$, close to the reference width $2l_0 = 15 \mu\text{m}$ for $l_0 = 7.5 \mu\text{m}$. This comparison is descriptive rather than an asserted analytical identity for the exact AT2 half-maximum profile. The mesh corridor remains wider than the observed damage band, so the regularized profile is resolved without forcing the crack path through a single row of elements.

9. Matched phase-field contours

Figure 7 provides the requested spatial-field check at three matched displacements. It compares the fixed 15,192-element reference and corrected 14,483-element ET1 adaptive mesh without using unmatched time frames.

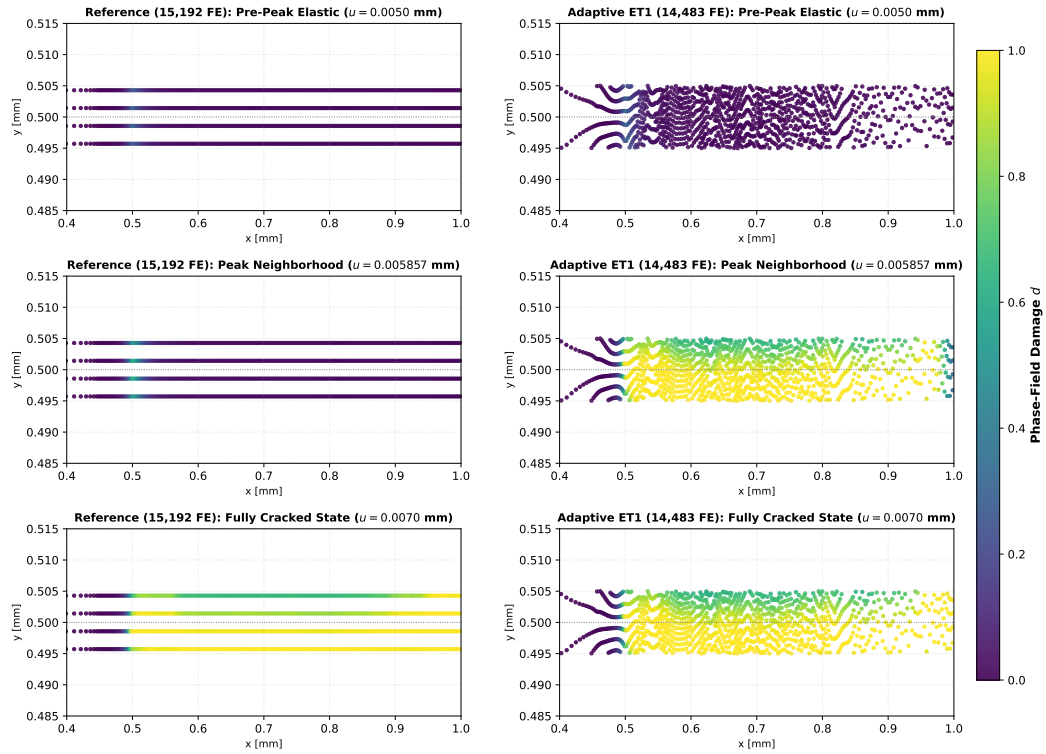


Figure 7: Matched phase-field damage contours for the fixed reference (left column) and corrected ET1 adaptive mesh (right column) at $u = 0.0050$ mm, the reference peak neighborhood $u = 0.005857$ mm, and the fully cracked state $u = 0.0070$ mm.

Scientific conclusion. The corrected adaptive solution localizes on the same horizontal ligament as the fixed reference and reaches complete specimen separation without off-axis crack growth.

The point patterns differ because the reference uses a structured corridor while the adaptive mesh is unstructured. The scientifically relevant comparison is therefore the field distribution and crack trajectory, not node-by-node visual identity.

10. Conclusions, limitations, and requested decision

10.1 Conclusions

1. The earlier native adaptive mesh was operational but spatially inefficient: the Step-1/pre-peak sizing field produced broad refinement.
2. Evaluating the localized Step-2 field corrected the mesh morphology and concentrated refinement along the horizontal crack-propagation ligament.
3. The preferred ET1 corrected mesh contains 14,483 finite elements, with 64.12% in the process corridor and no artificial refinement on the initial slit flanks.
4. The corrected mesh is qualitatively consistent with the localized pattern reported by Pandey and Kumar [2025]; exact reproduction of the published 13,941 count is not claimed.
5. Initial stiffness is preserved ($\Delta K_0 = -0.0261\%$ for ET1), and all Step-2 tolerance cases remain within the predeclared peak-force acceptance range.
6. Internal adaptive spatial convergence is demonstrated between 14.5k and 57.9k elements, with changes below 0.28% in $F_{\max}$ and u_{peak} .
7. Spatial damage profiles, symmetry, energy histories, and complete fine-mesh fracture evolution support the corrected adaptive workflow.

10.2 Limitations that remain explicit

- The persistent 2.13% peak-force offset between the spatial-fine adaptive mesh and the structured fixed reference is unresolved; internal adaptive convergence does not remove that discrepancy.
- The post-peak bookkeeping difference reaches 4.43% for the full-horizon spatial-fine run. This is convergence-consistent empirical evidence, not closure of an exact global energy identity.
- ET2–ET5 reduce cost but progressively under-resolve the process corridor; they should not replace ET1 when morphology fidelity is the primary objective.

Decision requested. Approve formal Gate-6B closure on the basis of the corrected localized mesh, stable global mechanics, quantified spatial convergence, and explicitly bounded energetic limitations. Only after that decision should work proceed to Gate 6C state-transfer verification. Mode-II and Gate 7 remain on hold.

References

- A. Pandey and S. Kumar. A simple and robust mesh refinement implementation in Abaqus for phase field modelling of brittle fracture. *Computer Modeling in Engineering & Sciences*, 144(3):3251–3286, 2025. doi: 10.32604/cmescs.2025.067858.